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Epidemiology of Injury and Illness in North American Professional Men's Soccer: Comparing COVID-19 Lockdown With Previous Seasons

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Background: After COVID-19 lockdown, studies across Europe and Asia examined its effect on professional soccer injury rates and severity; however, COVID-19 lockdown influence on injuries in United States men's professional soccer has not been evaluated.

Hypothesis: Injury and illness rates during the 2020 season were higher than the previous 2 seasons.

Study Design: Retrospective observational cohort study.

Level of Evidence: Level 3.

Methods: Medical staff from participating Major League Soccer (MLS) clubs entered injuries and athletic exposures during the study period into a league-wide electronic medical record system. Injury rate and severity were analyzed to examine differentials between the 2020 COVID season and historical controls.

Results: The injury incidence rate per 1000 hours in 2020 was 10.8, ie, higher than 2018 (5.4) and 2019 (5.0) ($P < 0.05$). Training injury incidence rate increased in 2020 (8.9) compared with 2018 (2.5) and 2019 (2.6) ($P < 0.05$), whereas match injury incidence decreased in 2020 (18.3) compared with 2018 (24.0) and 2019 (22.7) ($P < 0.05$). Incidence rates of lower extremity muscle injuries (6.04), anterior cruciate ligament (ACL) injuries (0.17), and concussions (0.49) were also higher in 2020 compared with 2018 (2.5, 0.07, 0.27) and 2019 (2.36, 0.05, 0.22) ($P < 0.05$). More injured players in 2020 missed >90 days (17.7%) than in 2018 (10.2%) and 2019 (10.1%) ($P < 0.05$). Incidence of all non-COVID-19 illness was higher in 2020 (3.93) than 2018 (1.53) and 2019 (1.32) ($P < 0.05$).

Conclusions: During 2020, there were significant increases in incidence rates of overall injuries, training injuries, lower extremity muscular injuries, ACL injuries, concussions, and non-COVID illness, along with a higher percentage of players missing >90 days compared with the 2 previous seasons.

Clinical Relevance: These results may help clarify the effects of future MLS inseason work stoppages and periods of restricted training.

Keywords: COVID-19; epidemiology; football (soccer); injury incidence

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On March 11, 2020, the World Health Organization (WHO) declared COVID-19 a global pandemic.[27] The impact was far-reaching, essentially shuttering almost every aspect of society. Professional sports leagues in the United States (US), including the National Football League (NFL), Major League Baseball (MLB), National Basketball Association (NBA), National Hockey League (NHL), and Major League Soccer (MLS), were shut down. MLS suspended all activity on March 12, 2020. The players were instructed to train in isolation until it was deemed safe by the league as well as local and regional jurisdictions to reconvene in small group settings.

Starting with the "MLS is Back (MIB) Tournament" on July 8, 2020, MLS was the first male professional sport league in the US to resume activity. In the setting of variable national, state, local, and league guidelines, clubs participated in differing levels of training before commencement of the tournament. Due to discrepancies in state regulations, 1 club was unable to begin training as a full group before their arrival to the MIB Tournament. All other teams completed a combination of single-person, small group (<5 players), and full team training sessions.

In addition to the evolving health-related concerns associated with COVID-19, questions were raised regarding the players' ability to withstand the physical and psychological demands of competition after the hiatus from activity.[16] The effects of undertraining, as well as rapid change in workload, are well-documented, specifically related to increased injury risk.[3,6,17,29] Before COVID-19, other professional sports leagues in the US have returned to play rapidly after imposed work stoppages, with increased injury rates.[13,22] Unlike preseason camp, where players report after completing a thorough off-season program, players resumed training before the MIB Tournament after an imposed hiatus in a significantly limited environment, where space, session duration and frequency, equipment, and human interactions were heavily regulated.

These same concerns regarding the hiatus from activity were shared across professional soccer leagues internationally, with multiple studies focusing on return-to-play protocols and recommendations for resumption of training.[2,3,21,23,35] Multiple studies across Europe and Asia also examined the effect of the COVID-19 lockdown on professional soccer injury rates, severity, and quality of play, with varying results.[1,5,10,14,18-20,25,26,28,31-33] However, the influence of the COVID-19 lockdown on injuries in the top division of US men's professional soccer has not been evaluated to date.

The primary objective of this study was to describe the epidemiology of injuries and illnesses sustained in professional men's soccer players in the US MLS during the MIB Tournament in 2020 after the COVID-19 playing stoppage and for the remainder of the regular 2020 season. In addition, this study aims to compare these results with injuries and illnesses sustained during the 2 previous historical reference seasons (2018 and 2019).

METHODS

Study Design and Participation

The study was approved in concept by Medical Assessment Research Committee (M-MARC) of the MLS, and Human Ethics Internal Review Board (IRB) approval was obtained from the University of Delaware, USA (IRB no. 1683181-1). Athletes from all clubs provided written consent at the beginning of each season or upon entry into the league. All MLS clubs (N = 26) participated fully in the study.

The study cohort consists of male professional soccer players in MLS with clubs in the US and Canada. This included 23 clubs in 2018, 24 clubs in 2019, and 26 clubs in 2020. If a player left and returned to an MLS club, his data were included in the study; however, any activity that was completed while away from MLS (ie, national team duty) was not available and not included. If a player left a club and did not return to MLS, his data were included up to time of departure.

Data Collection

Club medical staff (athletic trainers, physical therapists, and/or physicians) reported injuries and illnesses in real time through the league's mandatory electronic medical record (EMR) system. The system's standardized reporting form includes information including date and time of injury, injury type, mechanism of injury, severity of injury, weather conditions, and immediate medical treatment rendered. In addition to injuries and illnesses, staff recorded all sanctioned activities, in minutes, including training sessions, competitive and noncompetitive matches, and gym sessions. To ensure accuracy in reporting, each club's medical department was asked to report the following for 2018 to 2020: (1) preseason roster size, (2) regular season roster size, (3) average preseason training time per week, and (4) average regular season training time per week. Data from preseason 2018 through the end of the full 2020 MLS season were obtained directly from MLS by the investigative research group. Injuries were classified as "time loss injuries" if the player missed ≥1 training sessions or matches. A player was considered injured until he received full clearance by the club's medical staff and completed his first unrestricted practice session.[34]

Regular Schedule and COVID-19 Scheduling Disruptions

In a traditional MLS season, players report for preseason in January. Exact reporting dates are staggered depending on whether a club has qualified to participate in CONCACAF Champions League (CCL) or not, or on whether a player had additional national team requirements. For the purposes of this study, the reporting date is the day on which over two-thirds of teams in the league reported to preseason in a given season (Figure 1).

In 2018, the 34-game regular season began on March 3 and concluded on October 28. The league observed a 9-day break

Figure 1. 2018-2020 schedule characteristics.

in June for the 2018 FIFA Men's World Cup. MLS Cup Playoffs began on October 31 with the top 6 clubs from each of the 2 conferences (east and west) eligible to participate. The season concluded on December 8 with the MLS Cup Final.

In 2019, the 34-game regular season began on March 2 and concluded on October 6. MLS Cup Playoffs began on October 19 with the top 7 clubs from each of the 2 conferences eligible to participate under a new single-match elimination structure. The season concluded on November 10 with the MLS Cup Final.

In 2020, the regular season began on February 29. On March 12, all activity was suspended due to the COVID-19 pandemic. Clubs were eligible to begin individual outdoor training as their state and local governments allowed on May 6. MLS organized for each club to reconvene in Orlando, Florida, to participate in the MIB Tournament. This 1-site tournament allowed for a closed unit, without fans or outside visitors to mitigate COVID-19 transmission among players and staff. Testing for COVID-19 via polymerase chain reaction (PCR) occurred every other day throughout the tournament and other mitigation efforts were strictly enforced. The MIB Tournament began on July 8 and concluded on August 11. Clubs were eligible to arrive in Florida as early as June 24 but were required to arrive no later than 7 days before their first match. Two clubs withdrew from the tournament shortly after arriving secondary to a high percentage of players and/or staff testing positive for COVID-19. The remaining clubs played 3 games in the group stage of the tournament. These group stage games counted toward their regular season record and points total. After the conclusion of the Tournament, the regular season resumed August 12 and concluded November 8. Clubs played between 18 and 23 regular season games, including 2 games for most clubs before the COVID-19 play stoppage, the 3 group stage games during the MIB Tournament, and the matches played after the tournament. Playoffs ran from November 20 to the MLS Cup on December 12. Due to COVID-19 travel restrictions, after the MIB Tournament, Canadian-based MLS (N = 3) clubs played 6 total matches in Canada before arriving back in the US to set up temporary operations in a host city for the remainder of the season.

Statistical Analysis

Injury and illness time loss data were recorded for all clubs. Data for time loss injuries and illnesses were reported according to international reporting standards.[12,34] Injury and illness data for the full 2020 season (preseason, MIB, regular season, and postseason) were compared with the full 2018 and 2019 MLS seasons.

Athletic exposure hours were calculated by multiplying the number of rostered players by the average training time per week for the number of weeks during each respective timeframe, plus the result of the number of participating players multiplied by the average game time per week for the number of games during each respective timeframe.[24] Injury incidence rate was calculated by number of injuries divided by the total exposure hours and standardized by multiplying by 1000 hours.[24] Injury severity was defined by the required days to return to play, and categorized by the following time bins: 1 to 3 days, 4 to 7 days, 8 to 28 days, 29 to 90 days, 91 to 180 days, >180 days.[34] Incidence and proportions across 3 seasons were compared using the chi-square test for proportions. The significance level was defined as $P < 0.05$. All analyses were performed using R (Version 4.3.1, R Core Team 2021).

RESULTS

Overall Injuries, Exposure Hours, and Injury Incidence

In 2018, there were 891 injuries occurring over 168,513 hours with 152,954 training hours and 15,560 hours in matches. In 2019, there were 888 injuries occurring over 179,319 hours with 163,083 training hours and 16,236 hours in matches. In 2020, there were 712 injuries occurring over 65,695 hours with 52,342 training hours and 13,353 hours in matches. The total injury

Table 1. Number of teams, injuries, exposure hours, and injury incidence

	2018 Season	2019 Season	2020 Season	*P* Value
Number of teams	23	24	26	-
Number of injuries	891	888	712	-
Training exposure hours	15,2954	163,083	52,342	-
Match exposure hours	15,560	16,236	13,353	-
Total exposure hours	168,513	179,319	65,695	-
Total injury incidence per 1000 hours	5.4	5.0	10.8	<0.01*
Training injury incidence per 1000 hours	2.5	2.6	8.9	<0.01*
Match injury incidence per 1000 hours	24.0	22.7	18.3	<0.01*

*Significant at $P < 0.05$.

incidence rate in 2020 was 10.8, which was increased compared with the incidence rates in 2018 (5.4) and 2019 (5.0) (Table 1). The injury incidence during training was increased in 2020 (8.9) compared with 2018 (2.5) and 2019 (2.6), whereas the injury incidence during matches was decreased in 2020 (18.3) compared with the previous 2 seasons (24.0 and 22.7) (Table 1).

Injury Profile

Injuries were sorted into 15 different body regions (Table 2) according to *International Olympic Committee* standards.[12] The 3 most common regions regardless of season were thigh, knee, and ankle injuries (Table 2). There was a lower percentage of hip injuries in 2018 (5.7%), a lower percentage of groin injuries in 2019 (6.3%), and a higher percentage of thigh injuries in 2020 (35.7%) (Table 2). Injuries were also recorded according to type/diagnosis. In 2020, there were increased incidence rates of lower extremity muscular injuries (hip flexors, adductors, quads, hamstrings, gastrocnemius), ACL injuries, and concussions (Table 3).

Injury Severity

Since time loss data were right-skewed, the injury severity was calculated with medians.[34] In 2018, injured players missed a median of 17 days, 1 game, 5 practices, and 7 combined team events. In 2019, injured players missed a median of 19 days, 1 game, 5 practices, and 7 combined events. In 2020, injured players missed a median of 21.5 days, 1 game, 4 practices, and 6 combined events. The distribution of days missed according to time bins for each season is listed in Table 4. There was a higher percentage of players missing 4 to 7 days in 2018 (16%), along with higher percentages of players missing 91 to 180 days (9.6%) and >180 days (8.1%) in 2020 (Table 4).

Illness Incidence

Illnesses were categorized into 5 types: (1) common cold, general illness, or URI; (2) flu or flu-like symptoms; (3) all general medical; (4) all non-COVID-19; and (5) COVID-19 (Table 5). There were 257 total illnesses reported in 2018, 237 illnesses in 2019, and 387 illnesses in 2020. There were 129 COVID-19 cases in 2020. Illness incidence rates are reported in Table 5. There were higher incidence rates per 1000 hours of all general medical illness (3.25) and non-COVID-19 illness in 2020 (3.93) (Table 5).

DISCUSSION

Injury Incidence

The overall injury incidence rate per 1000 exposure hours in MLS was doubled during COVID-19 in 2020 (10.8) versus 2018 (5.4) and 2019 (5.0). Such an increase was also seen with COVID-19 in the Italian Serie A league and in Saudi Arabian soccer leagues.[1,20] In contrast, both the German Bundesliga and the Japanese Professional Football League had no difference in injury incidence.[14,19] One study examining UEFA (*Union of European Football Associations*) clubs across Europe reported increased injury incidence associated with training but no change in injury incidence associated with matches.[33] Our study similarly demonstrated increased injury incidence associated with fewer training sessions, and actually a decreased injury incidence associated with match play. The reasons for this variation in injury incidence are unclear; an argument for decreased or no change in injury incidence is that the lockdown allowed medical staff to rest and treat players with chronic overuse injuries.[14,33] Conversely, the limited access to training, coaches, and medical staff associated with the pandemic was theorized to lead to potential spikes in injuries.[9,10] The MLS has a different calendar of competition than other leagues, which could be a contributing factor, particularly in relation to the timing of the lockdown and subsequent return to play. Ultimately, in the setting of fluctuating incidence and prevalence of COVID-19, as well as national, local, and league regulations, variable effects of the pandemic on injury incidence are conceivable.

Table 2. Injury region (percentage of total injuries for that season)

	2018 Season	2019 Season	2020 (Season)	*P* Value
Head	6.7	6.0	5.6	0.63
Neck	0.6	0.7	0	0.99
Shoulder	1.9	2.0	0.7	0.07
Upper arm, elbow, forearm	0.9	0.7	0.3	0.30
Wrist and hand	0.8	0.5	1.1	0.31
Chest and thoracic spine	0.2	1.4	1.0	0.03*
Lumbosacral	3.7	3.3	2.2	0.24
Abdomen	1.5	2.0	0.8	0.15
Hip	5.7	9.7	10.3	<0.01*
Groin	9.7	6.3	8.3	0.03*
Thigh	30.3	31.0	35.7	0.05*
Knee	14.8	12.5	11.7	0.14
Lower leg	8.0	8.0	8.6	0.89
Ankle	10.9	10.7	10.0	0.83
Foot	4.4	5.4	3.8	0.29

*Significant at $P < 0.05$.

Table 3. Incidence of common injury types per 1000 hours

	2018 Season	2019 Season	2020 Season	*P* Value
Lower extremity muscle injuries	2.50	2.36	6.04	<0.01*
ACL injuries	0.07	0.05	0.17	0.01*
Concussions	0.27	0.22	0.49	<0.01*

ACL, anterior cruciate ligament.
*Significant at $P < 0.05$.

In addition to increased injury incidence overall, our study demonstrated increased incidence rates of lower extremity muscle injuries, ACL injuries, and concussions during the 2020 season. This increase in muscle injuries aligns with previous studies demonstrating that muscle injuries occur more often as a result of insufficient training or a condensed match schedule.[7,8,11,36] During the 2020 season, MLS players participated in only one-third of the expected training hours compared with 2018 (34%) and 2019 (32%). In addition, the Japanese and Italian professional leagues found increased incidence of muscle injuries during COVID-19.[19,20] The increased incidence of ACL injuries also aligns with previous evidence from Germany that short-term changes or abrupt increases in training intensity can facilitate severe knee injuries.[15]

Injury Severity

In this study, the majority of players missed between 8 and 28 days due to injury in 2020, which was unchanged from previous seasons and similar to other professional leagues globally.[14,20,33] However, there were a significantly greater percentage of players that missed between 91 and 180 days or >180 days in

Table 4. Injury Severity (percentage of players with injuries that season)

Injury Severity (days missed)	2018 Season	2019 Season	2020 Season	*P* Value
1-3	6.4	6.5	6.5	0.99
4-7	16.0	12.7	11.1	0.01*
8-28	47.1	46.6	42.1	0.09
29-90	20.2	24.0	22.6	0.15
91-180	6.7	5.5	9.6	<0.01*
>180	3.5	4.6	8.1	<0.01*

*Significant at $P < 0.05$.

Table 5. Illness incidence per 1000 hours

	2018 Season	2019 Season	2020 Season	*P* Value
Common cold, general illness, URI	0.35	0.33	0.53	0.08
Flu or flu-like symptoms	0.12	0.06	0.14	0.08
All general medical (undiagnosed, fever, upset stomach, etc)	1.05	0.93	3.25	<0.01*
All non-COVID-19 illness	1.53	1.32	3.93	<0.01*
COVID-19	0	0	1.97	-

URI, upper respiratory tract infection.
*Significant at $P < 0.05$.

2020. One possible explanation for this finding may be the increased incidence of ACL injuries. Another factor to consider is that professional athletes in the US were given the chance to opt out of their sport given safety concerns associated with the pandemic.[13] This decision to opt out or remain sidelined with an injury (due to COVID-19 safety concerns) that would have otherwise been cleared for play sooner could have influenced players decision to return to play post lockdown.

Illness Incidence

The illness incidence in the 2 seasons before the pandemic was similar to rates previously seen in European and Asian professional leagues.[4,30] In 2020, the non-COVID illness incidence rate increased over 2-fold (3.93) compared with 2018 (1.5) and 2019 (1.3). The likely reason for the increased incidence rate is increased screening and symptom reporting in the setting of the COVID-19 pandemic.

Limitations

This study had multiple limitations. Despite efforts to emphasize the reporting and standardization of methods, various reporters at each individual clubs were responsible for logging injuries, illnesses, and events. This variability may have led to differences in the reliability of reporting of injuries, illness, days missed, or categorization of injury types. Furthermore, we did not account for players exposed or infected with COVID-19 and how that may have affected injury incidence or severity. Last, the study was limited by its retrospective design and the inability to determine causality of the findings during the 2020 season.

CONCLUSION AND CLINICAL IMPLICATIONS

Training and match play cessation and restrictions in response to the COVID-19 pandemic created a novel situation never before seen in US professional soccer. Our study demonstrated multiple important findings associated with the 2020 season, including significant increases in incidence rates of injuries overall, injuries associated with training, lower extremity muscular injuries, ACL injuries, concussions, and non-COVID illness, along with increased severity of injury demonstrated by a higher percentage of players missing significant time compared with the 2 seasons before the pandemic. These results may be helpful in predicting the effects of future MLS

inseason work stoppages or periods of restricted training, and how to optimally and safely allow players to resume training and match play after an unforeseen schedule disruption. Future work could include investigating factors such as specific team or player training programs that may have been more successful in preventing injuries or improving recovery.

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